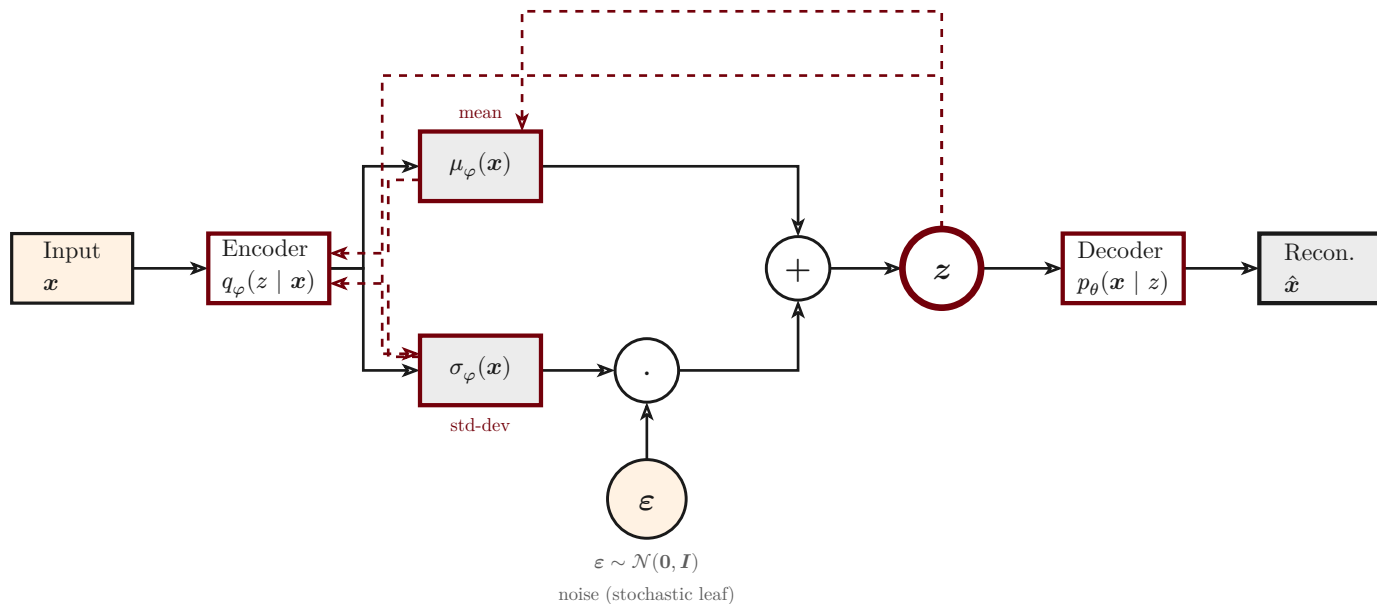


Reparameterization trick

$$z = \mu_{\varphi}(\mathbf{x}) + \sigma_{\varphi}(\mathbf{x}) \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$



—————➔ forward (data flow, deterministic)

- - - - -➔ backward (gradient of the ELBO)

Sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ directly is a **stochastic** node: no gradient can pass through a random draw. The trick moves the randomness into an auxiliary leaf $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and writes z as a **deterministic**, differentiable function of $(\mu, \sigma, \varepsilon)$. The backward pass (**dashed**) then flows through μ and σ into the encoder parameters φ — and simply **bypasses** the noise leaf ε .